

Image Classification Pipeline Report

Objective

The objective of this project is to create a reusable preprocessing pipeline for image datasets. The pipeline prepares images for machine learning by performing resizing, normalization, dataset splitting, and metadata generation.

Dataset Description

The dataset consists of images organized into folders. Each folder represents a class label.

Example:

- Cats
- Dogs

The folder names are used as class labels during preprocessing.

Preprocessing Steps

1. Image Loading

Images were loaded from the dataset directory using the Pillow library.

2. RGB Conversion

All images were converted to RGB format to ensure consistency across the dataset.

3. Image Resizing

Images were resized to 128×128 pixels.

Reason:

Machine learning models require all images to have identical dimensions.

4. Normalization

Pixel values originally ranged from 0 to 255.

Normalization was performed using:

$\text{Pixel Value} = \text{Pixel Value} / 255$

This scaled all values to the range 0–1.

5. Label Extraction

Folder names were used as class labels.

Examples:

- cats → Cats
- dogs → Dogs

6. Metadata Generation

A metadata file was created containing:

- Filename
- Label
- File Path
- Image Shape

The metadata was exported to metadata.csv.

7. Dataset Splitting

The dataset was divided into:

- 80% Training Data
- 20% Validation Data

using `train_test_split()`.

Results

The pipeline successfully:

- Loaded image data
- Standardized image dimensions
- Normalized pixel values
- Created metadata.csv
- Generated train and validation datasets

Applications

This preprocessing pipeline can be reused for:

- Image Classification
- Object Detection
- Face Recognition
- Medical Image Analysis
- Deep Learning Projects

Conclusion

The image preprocessing pipeline successfully automated image loading, resizing, normalization, metadata generation, and dataset splitting. The generated outputs are suitable for training machine learning and deep learning models.